

1. Unnamed Pipe (Parent to Child)

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>

int main() {
    int fd[2];
    pid_t pid;
    char write_msg[] = "Hello from parent!";
    char read_msg[100];

    if (pipe(fd) == -1) {
        perror("Pipe failed");
        exit(1);
    }

    pid = fork();

    if (pid < 0) {
        perror("Fork failed");
        exit(1);
    }

    if (pid > 0) {
        close(fd[0]);
        write(fd[1], write_msg, strlen(write_msg) + 1);
        close(fd[1]);
    } else {
        close(fd[1]);
        read(fd[0], read_msg, sizeof(read_msg));
        printf("Child received: %s\n", read_msg);
        close(fd[0]);
    }

    return 0;
}
```

2. Named Pipe (file1 to file2)

```
// file1.c (Sender)
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <sys/stat.h>
#include <unistd.h>
#include <string.h>

#define FIFO_NAME "myfifo"

int main() {
    int fd;
    char data[100];

    mkfifo(FIFO_NAME, 0666);

    fd = open(FIFO_NAME, O_WRONLY);

    printf("Enter message: ");
    fgets(data, sizeof(data), stdin);

    write(fd, data, strlen(data) + 1);

    close(fd);

    return 0;
}

// file2.c (Receiver)
#include <stdio.h>
#include <stdlib.h>
```

```

#include <fcntl.h>
#include <sys/stat.h>
#include <unistd.h>

#define FIFO_NAME "myfifo"

int main() {
    int fd;
    char data[100];

    fd = open(FIFO_NAME, O_RDONLY);

    read(fd, data, sizeof(data));

    printf("Received from file1: %s", data);

    close(fd);

    return 0;
}

```

3. Named Pipe (file2 to file1)

```

// file2.c (Sender)
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <sys/stat.h>
#include <unistd.h>
#include <string.h>

#define FIFO_NAME "myfifo"

int main() {
    int fd;
    char data[100];

    mkfifo(FIFO_NAME, 0666);

    fd = open(FIFO_NAME, O_WRONLY);

    printf("Enter message: ");
    fgets(data, sizeof(data), stdin);

    write(fd, data, strlen(data) + 1);

    close(fd);

    return 0;
}

// file1.c (Receiver)
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <sys/stat.h>
#include <unistd.h>

#define FIFO_NAME "myfifo"

int main() {
    int fd;
    char data[100];

    fd = open(FIFO_NAME, O_RDONLY);

    read(fd, data, sizeof(data));

    printf("Received from file2: %s", data);

    close(fd);

    return 0;
}

```

